

Coordinated Disclosure for Dual-Use AI in Protein Design

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In fall 2023, researchers at Microsoft Research discovered that open-source generative protein design tools could produce sequence variants of ricin, botulinum neurotoxin, and dozens of other dangerous proteins that evaded the biosecurity screening software used by DNA synthesis providers. Rather than publishing immediately, they spent ten months quietly developing patches, coordinating their distribution to synthesis companies worldwide, and only then releasing a full account in *Science* [1]. The process worked. But it worked because the team had the institutional resources and personal connections to orchestrate a discreet, months-long remediation effort without any formal protocol to guide or legitimize it. A researcher at a standard academic lab who makes the same discovery has no equivalent path: there is no committee to notify, no defined timeline, no agreed-upon criteria for assessing severity, and no mechanism for coordinating a response across the institutions that would need to act. In January 2024, David Baker and George Church published a landmark call in *Science* for mandatory screening and logging of all synthesized DNA sequences as a biosecurity control point [2]. Their proposal addresses what happens at synthesis endpoints. It does not address what happens computationally, the window between when a generative model produces a dangerous output and when anyone with authority to act knows about it. That window is the bottleneck.

The gap is not hypothetical from my position. I am completing a PhD in generative deep learning for protein design at INRIA, building models that predict and design protein-protein interactions at proteome scale [6]. The same architecture I use to identify therapeutic targets against bacterial pathogens can, with different training objectives, be directed toward binding interfaces that enhance virulence or evade immune recognition. RFDiffusion enables de novo design of proteins with defined structural properties and is publicly available [4]. AlphaFold3 predicts complex biomolecular interactions, including between proteins and nucleic acids, with accuracy that was not achievable three years ago [5]. In 2022, Urbina and colleagues demonstrated the dual-use floor of this trajectory by inverting the reward function of a generative drug design model and producing molecular candidates more toxic than VX nerve agent in under six hours, using publicly available tools [3]. These systems are now used by thousands of researchers worldwide, most of whom have no biosecurity training. Recognition of a dangerous output depends on the researcher having enough domain knowledge to understand what they are looking at. Reporting depends on there being somewhere to report to. Neither condition is reliably met.

The existing governance infrastructure was not built for this scenario. Institutional biosafety committees have authority over physical work with known pathogens and select agents but no mandate over computational outputs and no mechanism to evaluate sequences before a synthesis request is made. Export controls apply to specific listed agents and were written before AI could generate novel sequences with equivalent biological functions. Journals require dual-use research of concern statements, but these are applied after research is complete, depend on author self-assessment, and vary widely in scope across publications. The NSABB’s 2023 proposed biosecurity oversight framework explicitly acknowledged that AI-enabled biodesign tools are evolving faster than existing governance mechanisms can follow [7]. Baker and Church’s DNA screening proposal, consequential as it is, addresses the synthesis endpoint and leaves the upstream computational gap unaddressed. The field has overlapping post-hoc instruments and no prospective disclosure infrastructure for the moment a researcher’s model produces something dangerous.

The hypothesis is that a coordinated biological AI disclosure protocol, modeled on CVD in cybersecurity and on the informal process the Paraphrase Project demonstrated under pressure, can close this gap through a formal pre-established structure. The proposed pilot establishes a standing committee of 8 to 12 experts spanning biosecurity, structural biology, and AI-driven protein design, affiliated with three or four institutions that already have biosafety infrastructure. Researchers who discover potentially dangerous generative outputs submit a private report containing the sequences, the generation method, and a preliminary risk assessment. The committee has 90 days to evaluate the risk, consult domain experts, and recommend a response along a tiered spectrum ranging from unrestricted publication, to managed disclosure with sequence modifications, to indefinite suppression with notification to relevant national biosafety authorities. Decision criteria and response tiers are published before the first report arrives, so the committee is not constructing judgments ad hoc and researchers know what they are entering into. Unlike the Paraphrase Project's ten-month informal process, a standing committee with pre-agreed criteria and a defined intake path is accessible to any researcher regardless of institutional size or personal network. The pilot runs for two years with documented outcomes on every case.

The experiment has well-defined failure conditions that are each scientifically informative. If researchers do not submit reports, the committee's outreach effort documents whether the field perceives this risk as real or negligible; either finding shapes what intervention comes next and is worth recording. If reports arrive but the committee systematically disagrees on risk classification, the absence of shared criteria is itself a result pointing toward the next needed intervention: developing and validating risk assessment standards before scaling. If recommendations are produced but ignored, the diagnosis shifts from absent infrastructure to absent legitimacy, a different problem requiring a different fix. The Paraphrase Project showed that informal coordinated disclosure can work when the right people happen to be in the room. This pilot answers whether a formal, accessible version can work for researchers who are not.

The window for building this infrastructure proactively is closing. Baker and Church wrote in 2024 that the field had already crossed the threshold where deliberate misuse of protein design AI is a practical concern rather than a theoretical one [2]. The Paraphrase Project, published a year later, confirmed it empirically and showed that even an informal remediation required ten months. A formal protocol with a standing committee and pre-published criteria shortens that window and makes it accessible to researchers outside elite institutions. Once the first major incident attributable to AI-designed biological sequences arrives, governance will be built under crisis pressure by people who were not in the room when the research was done, shaped by whatever incident happened to be most visible rather than by a considered analysis of where the real risks lie. The author is completing a PhD in generative protein design at INRIA, building systems that cross this boundary operationally every week. Teaching AI fundamentals and governance to political science students at Sciences Po as an adjunct professor makes a second failure visible alongside the first: the future regulators in that classroom cannot evaluate whether a biosecurity proposal is technically sound or security theater, because their training has given them no vocabulary for it. The disclosure gap and the regulatory literacy gap reinforce each other, and a documented, functioning protocol is one of the few concrete objects that can anchor the education needed to close both.

References

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